

TELNET

Remote Interactive Computing

- **TCP can be used for interactive use of remote machines**
 - User establish remote login session
 - Then executes commands
- **Remote login may not be simple**
 - **Most systems designed to expect direct login**
 - May have to modify the machine's OS
 - **Building interactive client may be difficult, too**
 - Passing on Control-C commands

- **TELNET is not sophisticated as some**
 - Is widely available
- **ID remote machine by domain name or IP address**
- **Offers three basic services**
 - **Defines network virtual terminal**
 - Clients only have to build to the standard interface
 - **Allows negotiation of some standard options**
 - **Treats both ends of connection symmetrically**
 - Does not force keyboard input or screen output

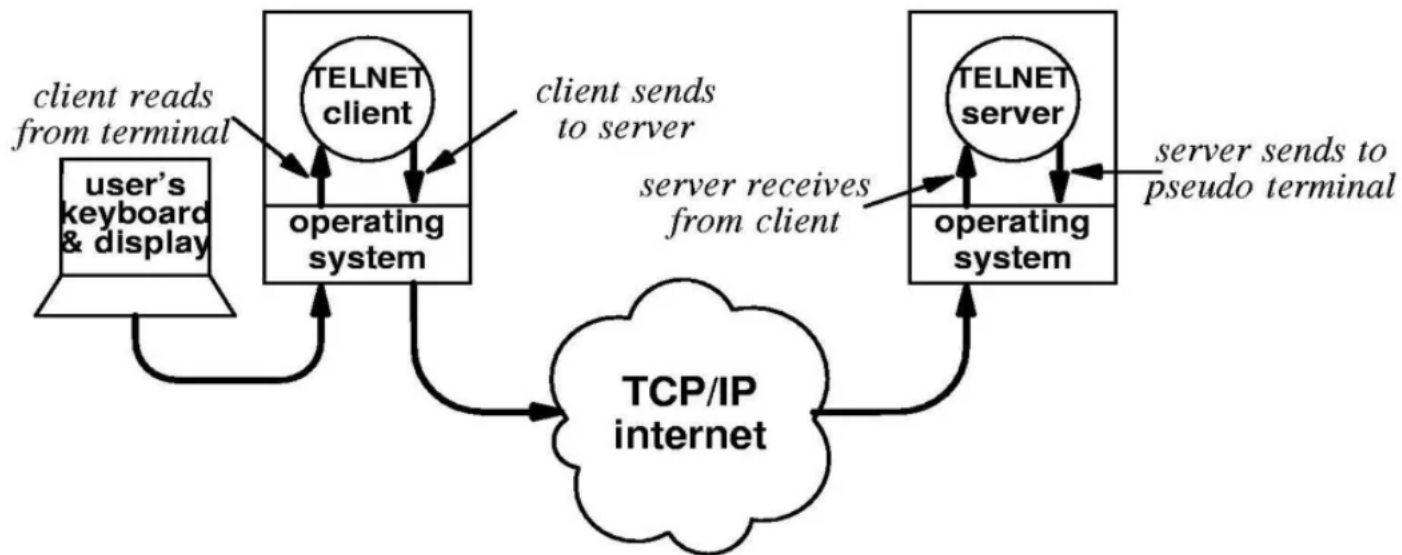


Figure 24.1 The path of data in a TELNET remote terminal session as it travels from the user's keyboard to the remote operating system. Adding a TELNET server to a timesharing system usually requires modifying the operating system.

TELNET Protocol

- **TCP/IP suite includes TELNET**

- **Simple remote terminal protocol**
- Allows user to log into computer over internet
 - Establishes TCP connection
 - Passes keystrokes to remote machine
 - Carries output back to user's screen
- **Transparent**
 - Gives appearance that user keyboard and display is directly connected to the remote machine

● **Server is more complex than shown**

- **Must handle multiple, concurrent connections**
- TELNET server consists of:
 - Master server that listens for new requests
 - Slave that handles one particular connection

● ***Pseudo terminal***

- **OS entry point**
- Allows application to transfer characters to OS as if they came from a keyboard
- Cannot build TELNET server without it

● Adv of TELNET server as application pgm

- **Modification & control of server easier**

- Since code is not part of OS

● Disadv

- **Inefficiency**

- Each keystroke

- User keyboard → OS → Client program → OS → Across internet
 - Then, Server's OS → Application Program

- Output comes back over same path

- Each keystroke requires several context switches

- Expensive; practical due to slow typing speeds